

# Bachelor AI - MLOps

## 05B INFRASTRUCTURE EXERCISE SETUP

Tobias Mérinat

`tobias.merinat@hslu.ch`



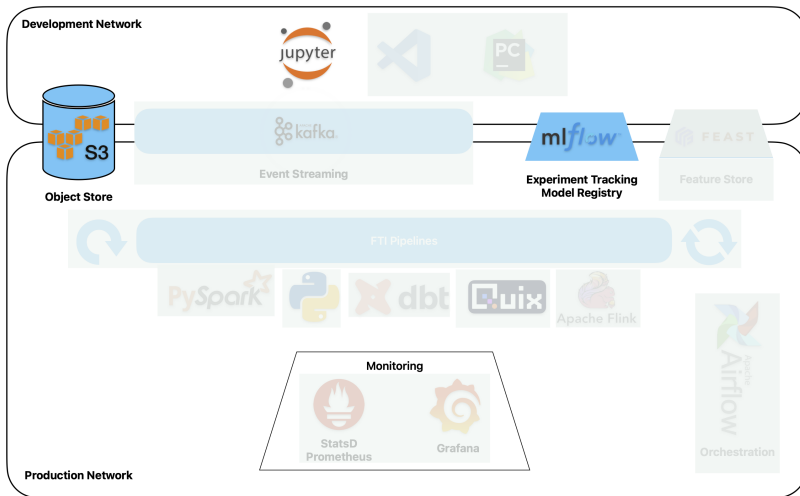
DEPARTMENT OF INFORMATION TECHNOLOGY  
Lucerne University of Applied Sciences and Arts  
6343 Rotkreuz, Switzerland

September 17/24 and October 15 2026

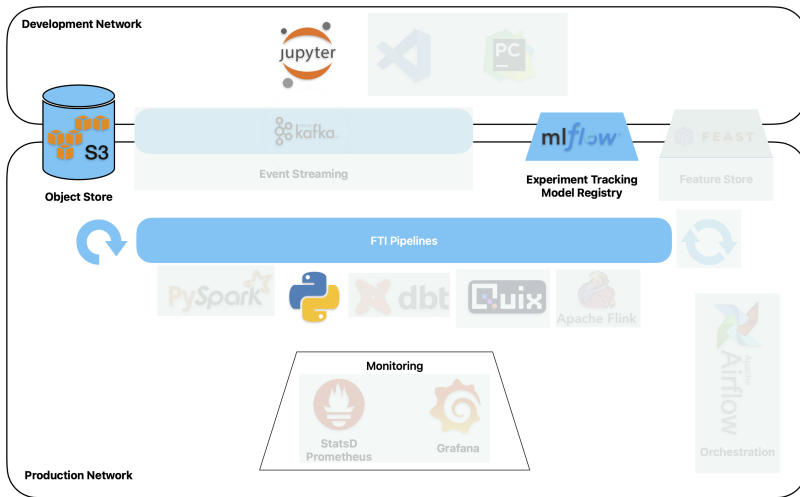
# Object Store and Development Environment



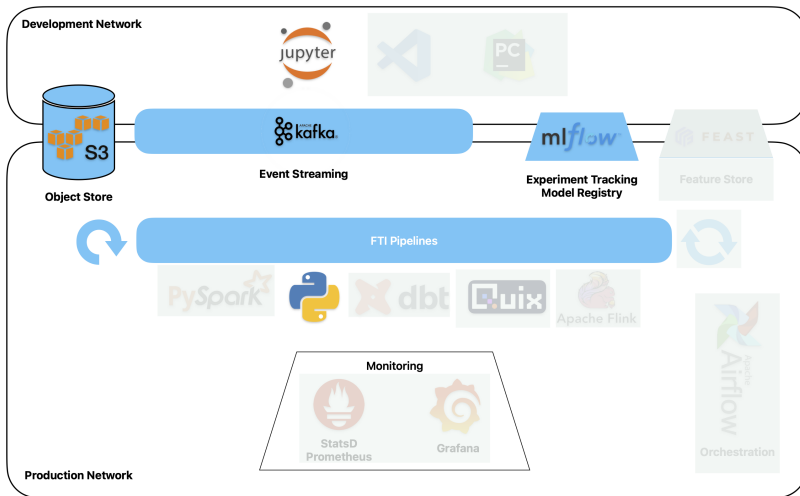
# Model Registry (MLFlow)



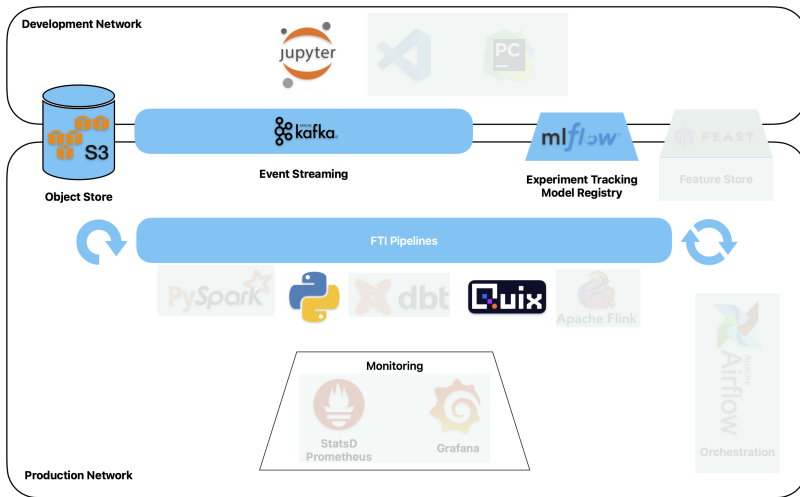
# Batch Pipeline with Python



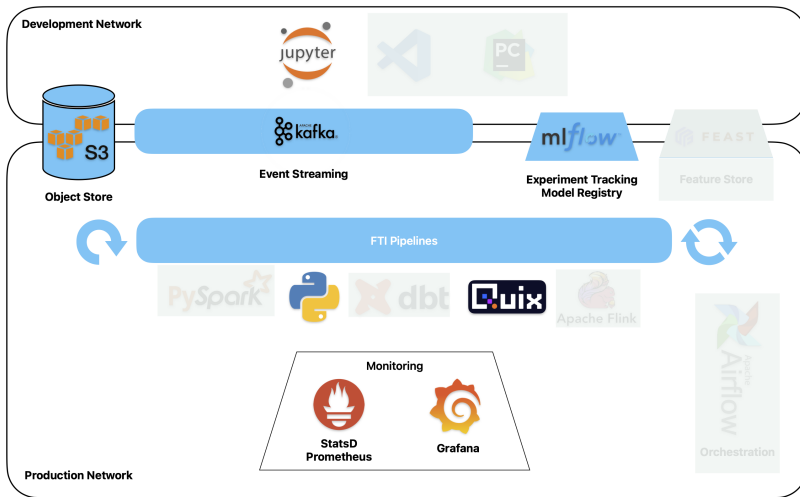
# Event Streaming Platform (Redpanda)



# Real time Pipeline with Quix



# Monitoring (Prometheus / StatsD / Grafana)



# Infrastructure is containerized

- All services run inside docker containers
- When you set up services, configure, start or stop them you are working on the host system (i.e. **not** inside a container)
- You **do not** build models or execute code on the host system, you always do this inside a container
  - In exercises that are about setting up infrastructure, you work outside of containers
  - In exercises where you build models, pipelines, run them, or access web frontends (jupyter, mlflow, grafana, . . . ) you work inside containers

- The following two containers are pre-configured:
  - Object Store (*objectstore/docker-compose.yml*)
  - Jupyter Development Environment (*development\_env/docker-compose.yml*)
- And these six will be set up by you in the course of the exercises:
  - MLFlow model registry and experiment tracking (*mlflow/docker-compose.yml*)
  - Redpanda (kafka-compatible) event streaming service and console (*event-streaming/docker-compose.yml*)
  - Prometheus, StatsD and Grafana for monitoring (*monitoring/docker-compose.yml*)
- For simplicity reasons, pipelines do not run in separate containers. But they should. . .



# Codespaces main window

